



Microcontrollers

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MSP430 Architecture

Memory Map

		Access
1FFFFh	Flash/ROM	Word/Byte
10000h		
0FFFFh	Interrupt Vector Table	Word/Byte
0FFE0h		
0FFDFh	Flash/ROM	Word/Byte
↕		
↕		
0200h	RAM	Word/Byte
01FFh		
	16-Bit Peripheral Modules	Word
0100h		
0FFh	8-Bit Peripheral Modules	Byte
010h		
0Fh	Special Function Registers	Byte
0h		

MSP430G2553 memory

Memory Organization

Table 8. Memory Organization

		MSP430G2153	MSP430G2253 MSP430G2213	MSP430G2353 MSP430G2313	MSP430G2453 MSP430G2413	MSP430G2553 MSP430G2513
Memory	Size	1kB	2kB	4kB	8kB	16kB
Main: interrupt vector	Flash	0xFFFF to 0xFFC0	0xFFFF to 0xFFC0	0xFFFF to 0xFFC0	0xFFFF to 0xFFC0	0xFFFF to 0xFFC0
Main: code memory	Flash	0xFFFF to 0xFC00	0xFFFF to 0xF800	0xFFFF to 0xF000	0xFFFF to 0xE000	0xFFFF to 0xC000
Information memory	Size	256 Byte	256 Byte	256 Byte	256 Byte	256 Byte
	Flash	010FFh to 01000h	010FFh to 01000h	010FFh to 01000h	010FFh to 01000h	010FFh to 01000h
RAM	Size	256 Byte	256 Byte	256 Byte	512 Byte	512 Byte
		0x02FF to 0x0200	0x02FF to 0x0200	0x02FF to 0x0200	0x03FF to 0x0200	0x03FF to 0x0200
Peripherals	16-bit	01FFh to 0100h	01FFh to 0100h	01FFh to 0100h	01FFh to 0100h	01FFh to 0100h
	8-bit	0FFh to 010h	0FFh to 010h	0FFh to 010h	0FFh to 010h	0FFh to 010h
	8-bit SFR	0Fh to 00h	0Fh to 00h	0Fh to 00h	0Fh to 00h	0Fh to 00h

Instruction set (part)

CMP

CMP[.W] ; CMP.B Compare source and destination

Syntax CMP src,dst or CMP.W src,dst; CMP.B src,dst

Operation dst + .NOT.src + 1 or (dst - src)

Description The source operand is subtracted from the destination operand. This is accomplished by adding the 1s complement of the source operand plus 1. The two operands are not affected and the result is not stored; only the status bits are affected.

Status Bits N: Set if result is negative (src > dst), reset if positive (src ≤ dst)
 Z: Set if result is zero (src = dst), reset otherwise (src ≠ dst)
 C: Set if there is a carry from the MSB of the result, reset otherwise
 V: Set if an arithmetic overflow occurs, otherwise reset

Instruction set (part)

*DECD[.W]

*DECD[.W] *DECD.B **Double-decrement destination**

Syntax	DECD dst or DECD.W dst; DECD.B dst
Operation	dst - 2 → dst
Emulation	SUB #2,dst; SUB.B #2,dst
Description	The destination operand is decremented by two. The original contents are lost.
Status Bits	N: Set if result is negative, reset if positive Z: Set if dst contained 2, reset otherwise C: Reset if dst contained 0 or 1, set otherwise V: Set if an arithmetic overflow occurs, otherwise reset. Set if initial value of destination was 08001 or 08000h, otherwise reset. Set if initial value of destination was 081 or 080h, otherwise reset.
Mode Bits	OSCOFF, CPUOFF, and GIE are not affected.

Instruction set (part)

***INCD[.W]**

***INCD[.W]; *INCD.B** Double-increment destination

Syntax INCD dst or INCD.W dst; INCD.B dst

Operation $\text{dst} + 2 \rightarrow \text{dst}$

Emulation ADD #2,dst; ADD.B #2,dst

Operation The destination operand is incremented by two. The original contents are lost.

Status Bits N: Set if result is negative, reset if positive

Z: Set if dst contained 0FFFEh, reset otherwise

Set if dst contained 0FEh, reset otherwise

C: Set if dst contained 0FFFEh or 0FFFFh, reset otherwise

Set if dst contained 0FEh or 0FFh, reset otherwise

V: Set if dst contained 07FFEh or 07FFFh, reset otherwise

Set if dst contained 07Eh or 07Fh, reset otherwise

Mode Bits OSCOFF, CPUOFF, and GIE are not affected.

Instruction set (part)

RLA

*RLA[.W]; *RLA.B Rotate left arithmetically

Syntax RLA dst or RLA.W dst; RLA.B dst

Operation **C <- MSB <- MSB-1 LSB+1 <- LSB <- 0**

Emulation ADD dst,dst; ADD.B dst,dst

Description The destination operand is shifted left one position. The MSB is shifted into the carry bit (C) and the LSB is filled with 0. The RLA instruction acts as a signed multiplication by 2.

An overflow occurs if **dst ≥ 04000h and dst < 0C000h** before operation is performed: the result has changed sign.

An overflow occurs if **dst ≥ 040h and dst < 0C0h** before the operation is performed: the result has changed sign

Instruction set (part)

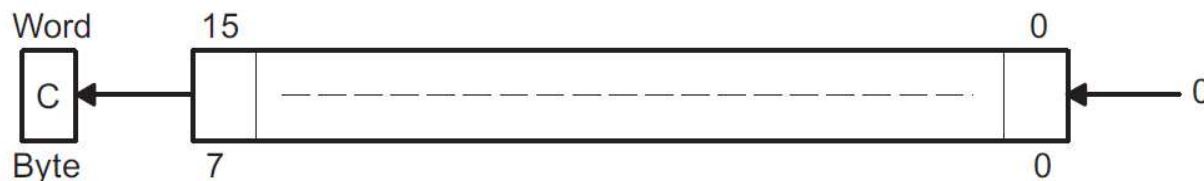
Status Bits	N: Set if result is negative, reset if positive Z: Set if result is zero, reset otherwise C: Loaded from the MSB V: Set if an arithmetic overflow occurs: the initial value is 04000h $\leq$ dst < 0C000h; reset otherwise Set if an arithmetic overflow occurs: the initial value is 040h $\leq$ dst < 0C0h; reset otherwise
Mode Bits	OSCOFF, CPUOFF, and GIE are not affected.

RLA Substitution

The assembler does not recognize the instruction:

RLA @R5+, RLA.B @R5+, or RLA(.B) @R5

It must be substituted by: ADD @R5+,-2(R5), ADD.B @R5+,-1(R5), or ADD(.B) @R5



Instruction set (part)

RLC

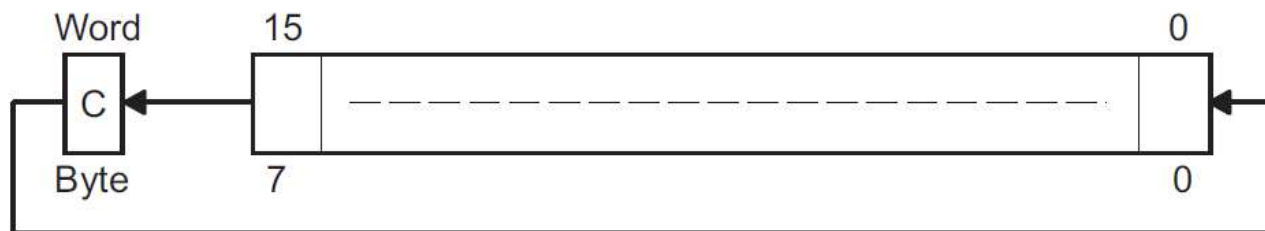
*RLC[.W]; *RLC.B Rotate left through carry

Syntax RLC dst or RLC.W dst; RLC.B dst

Operation $C \leftarrow \text{MSB} \leftarrow \text{MSB}-1 \dots \text{LSB}+1 \leftarrow \text{LSB} \leftarrow C$

Emulation ADDC dst,dst

Description The destination operand is shifted left one position. The carry bit (C) is shifted into the LSB and the MSB is shifted into the carry bit



Instruction set (part)

RLC

Status Bits	N: Set if result is negative, reset if positive Z: Set if result is zero, reset otherwise C: Loaded from the MSB V: Set if an arithmetic overflow occurs: the initial value is $04000h \leq dst < 0C000h$; reset otherwise Set if an arithmetic overflow occurs: the initial value is $040h \leq dst < 0C0h$; reset otherwise
Mode Bits	OSCOFF, CPUOFF, and GIE are not affected.

RLC and RLC.B Substitution

The assembler does not recognize the instruction:

RLC @R5+, RLC @R5, or RLC(.B) @R5

It must be substituted by: ADDC @R5+,-2(R5), ADDC.B @R5+,-1(R5),
or ADDC(.B) @R5

Instruction set (part)

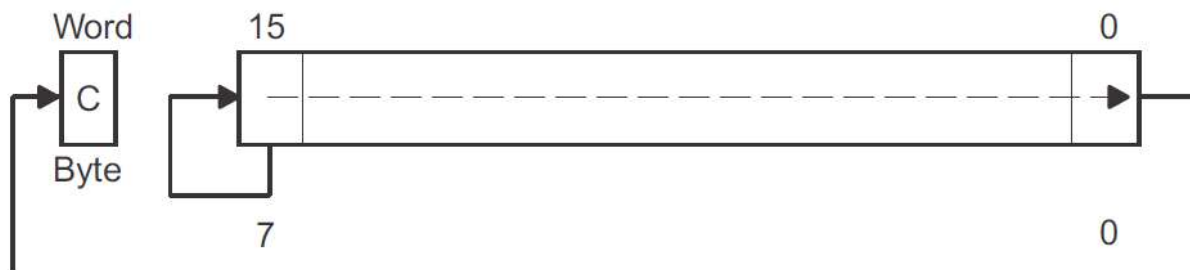
RRA

RRA[.W]; RRA.B Rotate right **arithmetically**

Syntax RRA dst or RRA.W dst; RRA.B dst

Operation **MSB $\rightarrow$ MSB, MSB $\rightarrow$ MSB-1, ... LSB+1 $\rightarrow$ LSB, LSB $\rightarrow$ C**

Description The destination operand is shifted right one position. The MSB is shifted into the MSB, the MSB is shifted into the MSB-1, and the LSB+1 is shifted into the LSB.



Instruction set (part)

RRA

Status Bits	N: Set if result is negative, reset if positive Z: Set if result is zero, reset otherwise C: Loaded from the LSB V: Reset
Mode Bits	OSCOFF, CPUOFF, and GIE are not affected.
RRA R5	; R5/2 -> R5 ; The value in R5 is to be multiplied by 0.75

Instruction set (part)

RRA

Status Bits N: Set if result is negative, reset if positive
 Z: Set if result is zero, reset otherwise
 C: Loaded from the LSB
 V: Reset

Mode Bits OSCOFF, CPUOFF, and GIE are not affected.

RRA R5 ; R5/2 -> R5
 ; The value in R5 is to be multiplied by 0.75 (0.5 + 0.25).

PUSH R5 ; Hold R5 temporarily using stack
 RRA R5 ; R5 x 0.5 -> R5
 ADD @SP+,R5 ; R5 x 0.5 + R5 = 1.5 x R5 -> R5
 RRA R5 ; (1.5 x R5) x 0.5 = 0.75 x R5 -> R5

Instruction set (part)

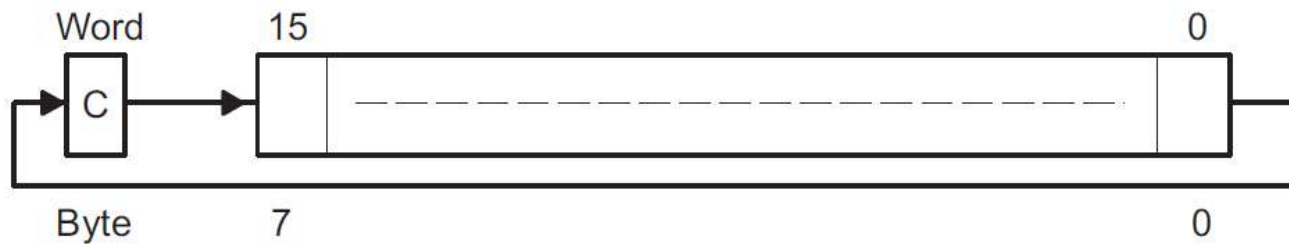
RRC

RRC[.W]; RRC.B Rotate right through carry

Syntax RRC dst or RRC.W dst; RRC dst

Operation $C \rightarrow \text{MSB} \rightarrow \text{MSB}-1 \dots \text{LSB}+1 \rightarrow \text{LSB} \rightarrow C$

Description The destination operand is shifted right one position. The carry bit (C) is shifted into the MSB, the LSB is shifted into the carry bit (C).



Instruction set (part)

RRC

Status Bits	N: Set if result is negative, reset if positive Z: Set if result is zero, reset otherwise C: Loaded from the LSB V: Reset
Mode Bits	OSCOFF, CPUOFF, and GIE are not affected.

SETC ; Prepare carry for MSB

RRC R5 ; R5/2 + 8000h -> R5

SETC ; Prepare carry for MSB

RRC.B R5 ; R5/2 + 80h -> R5; low byte of R5 is used

Instruction set (part)

SXT

SXT Extend Sign

Syntax

SXT dst

Operation

Bit 7 → Bit 8 Bit 15

Description

The sign of the low byte is extended into the high byte

Status Bits

N: Set if result is negative, reset if positive

Z: Set if result is zero, reset otherwise

C: Set if result is not zero, reset otherwise (.NOT. Zero)

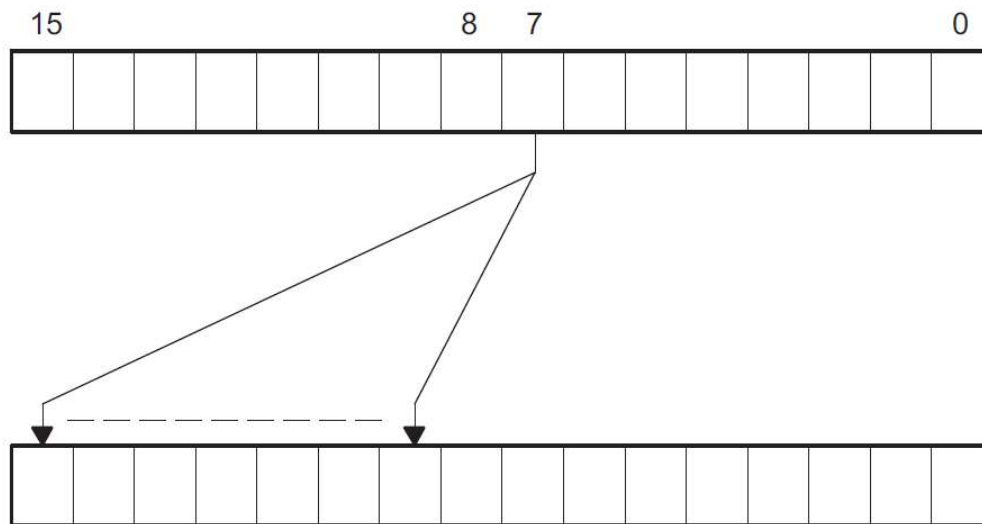
V: Reset

Mode Bits

OSCOFF, CPUOFF, and GIE are not affected.

Instruction set (part)

SXT





THANK YOU

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